

My Moon observation journal

Make eight observations over 3 to 4 weeks. Draw what you see before checking the phase guide.

Daytime safety

Use only your eyes and keep the Sun behind a building or tree. Never point binoculars, a telescope or a camera near the Sun.

Phase guide

This common Northern Hemisphere orientation may look rotated from your location. Follow the order, not left or right.



New Moon



Waxing crescent



First quarter



Waxing gibbous



Full Moon



Waning gibbous



Last quarter



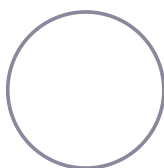
Waning crescent

Observation 1

Date: _____

Local time: _____

Weather: ☐ clear ☐ clouds



Direction: N E S W

Height: ☐ low ☐ middle ☐ high

My phase guess: _____

Growing / shrinking / not sure

Shade the part that looked dark

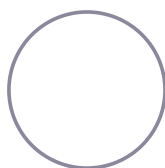
One more thing I noticed: _____

Observation 2

Date: _____

Local time: _____

Weather: ☐ clear ☐ clouds



Direction: N E S W

Height: ☐ low ☐ middle ☐ high

My phase guess: _____

Growing / shrinking / not sure

Shade the part that looked dark

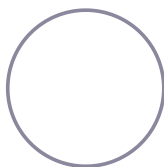
One more thing I noticed: _____

Observation 3

Date: _____

Local time: _____

Weather: ☐ clear ☐ clouds



Direction: N E S W

Height: ☐ low ☐ middle ☐ high

My phase guess: _____

Growing / shrinking / not sure

Shade the part that looked dark

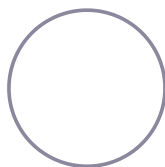
One more thing I noticed: _____

Observation 4

Date: _____

Local time: _____

Weather: ☐ clear ☐ clouds



Direction: N E S W

Height: ☐ low ☐ middle ☐ high

My phase guess: _____

Growing / shrinking / not sure

Shade the part that looked dark

One more thing I noticed: _____

My observing place and extra sky notes

My Moon observation journal

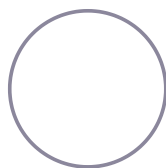
STEM Little Explorers

Observation 5

Date: _____

Local time: _____

Weather: ☐ clear ☐ clouds



Direction: N E S W

Height: ☐ low ☐ middle ☐ high

My phase guess: _____

Growing / shrinking / not sure

Shade the part that looked dark

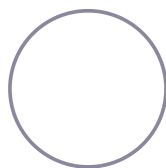
One more thing I noticed: _____

Observation 6

Date: _____

Local time: _____

Weather: ☐ clear ☐ clouds



Direction: N E S W

Height: ☐ low ☐ middle ☐ high

My phase guess: _____

Growing / shrinking / not sure

Shade the part that looked dark

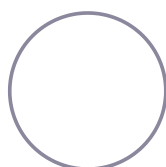
One more thing I noticed: _____

Observation 7

Date: _____

Local time: _____

Weather: ☐ clear ☐ clouds



Direction: N E S W

Height: ☐ low ☐ middle ☐ high

My phase guess: _____

Growing / shrinking / not sure

Shade the part that looked dark

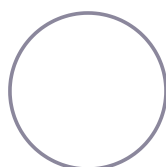
One more thing I noticed: _____

Observation 8

Date: _____

Local time: _____

Weather: ☐ clear ☐ clouds



Direction: N E S W

Height: ☐ low ☐ middle ☐ high

My phase guess: _____

Growing / shrinking / not sure

Shade the part that looked dark

One more thing I noticed: _____

Look back at your observations

Use your drawings as evidence. It is fine to revise your first idea.

1. What pattern did the illuminated shape follow?

3. What evidence shows that Earth's shadow does not cause the ordinary phases?

2. Did the Moon appear in the same part of the sky each time? What changed?

4. My first prediction was... Now I think...
